BEFORE YOU READ

The properties on this page

Each property describes one facet of a function: its symmetry, how it maps inputs, its direction, its bounds, its extremes, its continuity and its shape.

- L0 **Function properties**
  - L1 **Symmetry**
    - L2 **Even** mirror in the y-axis $f(-x) = f(x)$
    - L2 **Odd** point symmetry $f(-x) = -f(x)$
  - L1 **Mapping**
    - L2 **One-to-one** horizontal line test
    - L2 **Onto** range equals codomain
    - L2 **Bijective** both
  - L1 **Monotonicity**
    - L2 **Increasing or decreasing** on an interval (a, b)
    - L2 **Monotonic** one direction throughout
  - L1 **Bounded** outputs within [m, M]
  - L1 **Extrema**
    - L2 **Local**
    - L2 **Absolute**
  - L1 **Continuity and limits**
    - L2 **Continuous**
    - L2 **Asymptote**
  - L1 **Periodic** $f(x + p) = f(x)$
  - L1 **Zero at c** $f(c) = 0$
  - L1 **Concavity** up or down

TAKEAWAY

Check each facet in turn

Describing a function fully means walking through these facets one by one; each has its own section above.